ELS_Interface — User Guide

ELS_Interface is an interface board (a *shield*) that plugs into the **NUCLEO-G474RE** development board from STMicroelectronics. It connects the microcontroller to everything that lives "outside": the 12 V supply, a display, the encoders, the command buttons, the isolated outputs for the machine and a memory that keeps its data when the board is powered off.

This guide is written for whoever assembles, connects and uses the board. It explains where the supply goes, what every connector is for, how the indicator LEDs are used and which settings must be made on the Nucleo board. It does not describe the software.

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1. Power supply

1.1 Two ways to power the system

The board can work in two configurations.

"Machine" configuration — 12 V

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   12 V power supply          ELS_Interface board                Nucleo
  ──────────────────          ──────────────────────             ──────
+12 V ──► J3 pin 1 ──► F1 ──► D1 ──► U1 (buck 5 V) ──► E5V ──► CN7-6
GND   ──► J3 pin 2 ──► common ground
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"Bench" configuration — USB cable only

The Nucleo is powered by the computer through its USB cable. The board takes its 5 V from that cable (protected by diode D5) and works the same way, with no 12 V connected.

1.2 Protection on the 12 V input

Component	What it is	What it does
F1	1.1 A polyfuse	if something short-circuits, it interrupts the current and then resets itself
D1	protection diode (TVS) SMBJ15A	clips voltage spikes above about 24 V
C1	electrolytic capacitor 100 uF / 35 V	smooths the input voltage

Component	What it is	What it does
C3	capacitor 22 uF / 25 V	fast filter close to the converter

> **Warning: C1 and C5 are polarised.** The positive terminal (marked `+` on the board) must face the right side, otherwise the capacitor is damaged and may short the board.

1.3 From 12 V to 5 V: the converter U1

The heart of the supply is **U1**, a TPS5430 *buck* converter: it takes the 12 V, chops them at high frequency and produces a stable **5.00 V** with up to 3 A.

Component	Value	What it does
U1	TPS5430	the converter itself
L1	22 uH inductor	stores energy on every cycle
D2	SS34 diode	carries the current during the "off" phase
R1 / R2	10 kOhm / 3.24 kOhm	set the output voltage to **5.00 V**
C5	220 uF / 6.3 V tantalum	smooths the output, **watch the polarity**
C15	100 nF	filters high-frequency noise

The converter output is called **E5V** and goes straight to pin **CN7-6** (J1-6) of the Nucleo: that is what powers the microcontroller.

1.4 The board 5 V rail: D4 and D5

Besides the 5 V for the Nucleo, the board has a second 5 V rail called **V_5V_SHIELD**, which powers the logic display, the level shifters and the isolated outputs. This rail can receive current from **two different sources**, and diodes D4 and D5 act as one-way valves: they let the current in and stop it from going back.

Diode	Lets current flow from	To	Works when
D4	E5V (the converter 5 V)	V_5V_SHIELD	the 12 V is connected
D5	the Nucleo USB cable	V_5V_SHIELD	the board runs on USB only

The result is a rail at **4.4–4.7 V** (the difference from 5 V is the diode drop). Every part on the board accepts this value.

> **Warning: never remove D5.** Without that diode the board 5 V would flow back into the USB port of the computer.

1.5 Display supply

The display needs 12 V. Its branch is protected by **F2** (0.5 A polyfuse) and **D3** (SS34 diode), with **C6** (10 uF) as a filter. If F2 trips, the rest of the board keeps working: only the display goes dark.

1.6 Voltages you should find

Measurement point	Expected value	Where you measure
12 V input	12.0 V	J3 pin 1
Converter output **E5V**	**5.00 V +/- 0.15 V**	L1 pad 2
Board rail **V_5V_SHIELD**	**4.4–4.7 V**	C12 pad 1
Display 12 V	12.0 V	J7 pin 1
3.3 V (from the Nucleo)	3.30 V	J1 pin 12

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2. Connectors

2.1 J3 — 12 V supply input

Two-pole screw terminal, 3.5 mm pitch.

Pin	Signal	What it is for
1	+12 V	the positive wire of the supply
2	GND	ground (the negative wire)

2.2 J1 and J2 — the two sockets for the Nucleo

These are the sockets that hold the Nucleo board (morpho headers, 2x19 pins, 2.54 mm pitch). They are mounted on the **bottom** side of the board. Only a few pins are actually used by the shield:

Socket	Pin	Signal	What it is for
J1 (CN7)	**6**	E5V	brings the 5 V to the Nucleo: this is the MCU supply
J1	**28**	PA0	lights LED1
J1	**30**	PA1	lights LED2
J1	2	PC11	SDA data line of the EEPROM memory
J1	12, 16	3.3 V	reference supply and pull-up resistors
J1	8, 19, 20, 22	GND	ground
J2 (CN10)	**8**	USB VBUS	the 5 V of the USB cable (input of diode D5)
J2	2	PC8	SCL clock line of the EEPROM memory
J2	9, 20	GND	ground

> **Warning:** before inserting the Nucleo check that **pin 1** of J1 and J2 is aligned. Always insert the board with the power off.

2.3 J5 — field inputs and outputs

Ribbon (IDC) connector, 2x10 pins. Encoders, pushbuttons and the emergency stop are connected here.

Pin	Signal	What it is for
1	5 V (board rail)	supply for external devices
2	GND	spindle encoder ground
3	PC0 — spindle encoder A	first channel of the spindle encoder
4	PC1 — spindle encoder B	second channel of the spindle encoder
5	PC4	free auxiliary signal
6	GND	jog encoder ground

Pin	Signal	What it is for
7	5 V (board rail)	second supply output
8	PA6 — jog encoder A	first channel of the manual (jog) encoder
9	PA4 — jog encoder B	second channel of the jog encoder
10	GND	pushbutton ground
11	PB7 — EMERGENCY	the emergency stop (closed to ground)
12	—	not connected
13	PB10 — RESET button	reset pushbutton
14	PB12 — MODE button	mode selection pushbutton
15	PB14 — JOG + button	manual feed pushbutton (plus)
16	PA15 — JOG − button	manual feed pushbutton (minus)
17	GND	ground
18–20	—	not connected

All these signals pass through the level shifter **U2 (TXB0104)**, which adapts the 3.3 V of the microcontroller to the 5 V of the outside world.

2.4 J7 — serial display

8-pole connector, 2.00 mm pitch, for the DWIN display.

Pin	Signal	What it is for
1, 2	display 12 V	display supply (pin 2 in parallel to carry more current)
3	—	not connected
4	DWIN_RX	data coming **from** the display to the board (through R4, 100 Ohm)
5	DWIN_TX	data going **from** the board to the display
6	—	not connected
7, 8	GND	ground

2.5 J8 — isolated outputs

5-pole terminal. The first three outputs pass through an optocoupler (PC817): the signal crosses a layer of light, so the board and the machine are **electrically separated** — a surge in the machine cannot damage the microcontroller.

Pin	Signal	What it is for
1	U3 opto output	load driven by optocoupler 1
2	U4 opto output	load driven by optocoupler 2
3	U5 opto output	load driven by optocoupler 3
4	PC9	direct digital output, **not** isolated
5	GND	common ground for the loads

The load must be supplied externally and returned to the output pin.

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3. Debug LEDs

The board has two LEDs that show what the system is doing without any instrument.

LED	Colour	MCU pin	Resistor	Current
LED1	green	**PA0** (from J1-28)	R19 = 47 Ohm	4–18 mA
LED2	red	**PA1** (from J1-30)	R18 = 100 Ohm	7–15 mA

How they work: the microcontroller pin goes **high** (3.3 V), the current flows through the resistor, lights the LED and returns to ground. A **low** pin (0 V) means the LED is off.

Practical examples:

- To light the green LED, drive **PA0** high.
- To light the red LED, drive **PA1** high.
- To make them blink, alternate high and low with a delay.
- If an LED stays dark, check the polarity: the **cathode** (the side with the notch) goes to ground, the **anode** towards the resistor.

> **Note:** the green LED has a high forward voltage (2.6–3.1 V) and stays clearly visible with 47 Ohm. If more light is needed it can go down to 33 Ohm, without exceeding 25 mA.

The Nucleo also carries a third LED (LD2, connected to PA5): that one belongs to the Nucleo board, not to the shield.

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4. Jumper on the Nucleo: JP5

The **Nucleo** board (not the shield) carries a 4x2 jumper header **JP5** that decides **where the Nucleo takes its 5 V from**. It is the only manual setting of the system.

JP5 position	5 V source	When to use it
1-2	Nucleo USB cable	bench use, USB only
3-4	VIN	supply from the VIN pin
5-6	**E5V (from the ELS_Interface board)**	**normal use with the 12 V connected**
7-8	VBUS direct	special use, not recommended

Practical example: you are powering the board from 12 V and want the Nucleo powered too, so set JP5 on pins **5-6** (the Nucleo silkscreen says "E5V").

Practical example: you are at the bench with only the USB cable and no 12 V, so set JP5 on pins **1-2** ("5V_STLK").

With JP5 on 5-6 the programming section of the Nucleo stays powered from the shield, so the microcontroller can be programmed **even with the 12 V connected**.

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5. Things you must not do

> **1. Do not power from 12 V with JP5 on 1-2 or 7-8.** You would put two 5 V sources on the same MCU rail. The jumper must be on **5-6**.

> **2. Do not remove D5.** It is the diode that stops the board 5 V from entering the computer USB port.

> **3. Do not exceed 500 mA on the E5V pin (CN7-6).** That is the limit of the regulator inside the Nucleo.

> **4. Respect the polarity of C1 and C5.** A reversed electrolytic or tantalum capacitor is damaged and may short.

> **5. Do not bypass F1, F2, D1, D3.** They are the protection of the board and of the Nucleo: they matter exactly when something goes wrong.

> **6. Do not connect inductive loads (relays, motors, solenoid valves) to J5.** Loads belong on the isolated outputs of **J8**.

> **7. Always insert the Nucleo with the board powered off,** checking that pin 1 is aligned.

> **8. The 12 V supply must be EN 62368-1 certified.** It is an ST requirement for the E5V input.

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6. Design verification

The board described in this guide was checked with the `kicad-mcp-pro` MCP server (KiCad 10.0.6) and with `kicad-cli`:

Check	Result
ERC — electrical rules check of the schematic	**0 violations**
DRC — design rules check of the PCB	**0 active violations**, **0 unconnected items**
Netlist	exported correctly, **56 components**
Schematic / PCB parity	**56 = 56**, no missing component
3D models	**56 of 56** resolved